Ishita P Acharya

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EDUCATION

Mangalore Institute of Technology and Engineering (MITE) Nov 2022 – September 2026 (Expected) *Bachelor of Engineering in Computer Science - (Current GPA 7.69 / 10.0) Moodabidre, Karnataka, India*

Relevant Coursework: Data Structures, Algorithms, Operating Systems, Computer Networks, DBMS, Cloud Computing, Parallel Computing, Artificial Intelligence, Machine Learning, Software Engineering, Cryptography and Network Security.

ACADEMIC PROJECTS

Peripheral Arterial Disease Detection Using Machine Learning **March 2025 – November 2025**
Team Project (4)

* Built an end-to-end ML pipeline to detect Peripheral Arterial Disease (PAD) from unstructured Doppler reports by integrating OCR (Tesseract) with feature extraction and classification models.
* Developed a Random Forest–based classification system with SMOTE to address class imbalance, with model accuracy of 75% and enhancing recall for minority classes.
* Built a scalable Flask-based backend pipeline integrating OCR, ML inference, and rule-based validation for automated clinical report processing.
* Deployed a Flask-based web application for real-time report analysis, enabling automated extraction, prediction, and clinically interpretable results.

Voice-Controlled Music Player **November 2025 – January 2026**
Personal Project

* Developed a real-time voice-controlled music system enabling hands-free playback control using speech commands.
* Implemented speech-to-text and text-to-speech pipelines using SpeechRecognition for accurate command recognition and natural audio feedback.
* Integrated Spotify API for remote music control, supporting actions such as play, pause, track navigation, and volume adjustment.
* Designed a modular backend system using Python and Flask to process voice input, execute commands, and ensure low-latency response.

ClientPulse-Client Project & Delivery Intelligence Platform **April 2026 – Present**
Personal Project

* Developed and secured RESTful APIs using JWT-based authentication and role-based access control for Admin, Project Manager, Employee, and Client modules, ensuring efficient user management and secure data handling across the platform.
* Built and optimized automated invoice generation workflows by integrating approved employee timesheets, reducing manual processing effort and improving billing accuracy and operational efficiency.
* Designed and implemented scalable MongoDB database schemas to support multiple billing models, including Fixed Cost and Retainer-based systems, enabling flexible project and financial management.
* Improved backend system performance by achieving sub-500ms API response times and maintaining over 70% test coverage using Jest, ensuring reliable, efficient, and well-tested application functionality.

SKILLS AND CERTIFICATIONS

Programming Languages: C, Python
Frontend: React.js, HTML5, CSS3, Bootstrap
Backend: Node.js, Flask, FastAPI, REST APIs.
AI/ML: Scikit-learn, PyTorch (or TensorFlow), RAG, Langchain, OpenCV, Tesseract OCR.
Databases: MySQL, MongoDB.
Cloud, DevOps & Systems: AWS EC2, S3, Ubuntu, Linux, Git, GitHub, Railway.
Certifications: Cyber Job Simulation (Delloite), Prompt Wars (MITE), Cybersecurity in the Product Development Lifecycle: From Design to Deployment (MITE), Cybersecurity: Vulnerability Assessment and Exploitation (MITE).